

Exhaust Gas Back Pressure Report

Report Generated: 2026-06-07 14:25:06

System Parameters

Mass Flow	80.651 kg/s
Temperature	240 °C
Pipe Roughness	Steel Welded
Max Allowed BP	3000 Pa

Overall Assessment

Total Back Pressure	5069.26 Pa
Total Back Pressure	516.921 mmWC
Compliance Status	FAILED
Safety Margin	-68.98%

Pipeline Elements Breakdown

#	Element	Velocity (m/s)	ξ	ΔP (Pa)
1	Pipe	58.30	0.007	8.3
2	Pipe Bend	58.30	0.377	440.7
3	Custom	58.30	1.000	1169.3
4	Wye Branch	58.30	0.200	233.9
5	Diffuser	58.30	0.313	0.0
6	Butterfly Valve	58.30	0.332	388.1
7	Gate Valve	58.30	0.069	81.0
8	Silencer	58.30	2.350	2747.9